THREADFLOW

Technical Design Document — Knitwear Production Tracking & Intelligence System

PROJECT GITHUB REPOSITORY:
https://github.com/AgnivaSardar/ThreadFlow

LIVE WEB APPLICATION:
https://threadflow-ptun.onrender.com

BACKEND PRODUCTION API:
https://threadflow-backend-z7s9.onrender.com

TARGET DEPLOYMENT TOPOLOGY:
Render Static + Node Web Service, Supabase PostgreSQL 16

CLIENT & INDUSTRY CONTEXT:
Company A — Knitwear Exporter, Tiruppur | Version: 2.1 (Continuous Academic Edition)

TARGET HARDWARE PROFILE:
Android 10+ (2–3 GB RAM Mobile Devices), Desktop Modern Browsers

1. CLIENT PROBLEM, BASELINE ASSUMPTIONS & PRODUCT RESPONSE

Current Operating Environment: Company A coordinates export purchase orders across three owned production units and subcontracted printing and washing partners in the Tiruppur knitwear cluster. Order status historically existed in disconnected spreadsheets, separate time-and-action (T&A) plans, telephone follow-ups to floor supervisors, and buyer status inquiries handled via email and messaging. These fragmented communication channels prevented factory leadership from detecting production shortfalls until shipments were already irrecoverably delayed.

1.1 Formal Problem Statement Assumptions Register

All financial metrics, capacity models, and business justifications in this document derive strictly from the authoritative parameters specified in the problem statement. Every assumption has been formally tagged and labeled below:

ASSUMPTION TAG	DOMAIN PARAMETER	QUANTIFIED VALUE	OPERATIONAL & ECONOMIC DERIVATION
[ASSUMPTION 1]	Annual Export Order Volume & Value	500 orders / year at ₹21 lakh average	Factory manufactures 500 purchase orders annually with an average contract value of ₹21 lakh, establishing an annualized baseline turnover of ₹105 Crore.
[ASSUMPTION 2]	Baseline Late Shipment Rate	8% of total orders (40 orders / year)	Historically, exactly 8 percent of all purchase orders (40 orders annually) exceed contractual ex-factory dates due to delayed floor visibility.
[ASSUMPTION 3]	Emergency Air Freight Penalty	15 orders at ₹3.5 lakh extra = ₹52.5 lakh	Of the 40 delayed orders, 15 require emergency air freight conversion instead of sea freight to prevent retail cancellation, incurring ₹52.5 lakh in unrecoverable freight surcharges.
[ASSUMPTION 4]	Buyer Late Delivery Discount Penalty	25 orders at 5% discount = ₹26.25 lakh	The remaining 25 delayed orders incur a mandatory 5 percent contractual penalty discount ($25 \times (5\% \times ₹21 \text{ lakh}) = 25 \times ₹1.05 \text{ lakh} = ₹26.25 \text{ lakh}$) deducted from invoice payouts.
[ASSUMPTION 5]	Merchandiser Status-Chasing Overhead	12 merchandisers × 2 hrs/day = 7,200 hrs/yr at ₹400 = ₹28.8 lakh	Twelve export merchandisers spend 2 hours daily chasing status via telephone ($12 \times 2 \times 300 \text{ working days} = 7,200 \text{ hours/year}$). At ₹400/hr loaded cost, this drains ₹28.8 lakh in labor.
[ASSUMPTION 6]	Total Annual Operational Bleed	₹1.08 Crore / year (₹1,08,00,000)	Cumulative baseline bleed equals $₹52.5\text{L (air freight)} + ₹26.25\text{L (discounts)} + ₹28.8\text{L (labor)} = ₹1.08 \text{ Crore}$ annually.
[ASSUMPTION 7]	Implementation Budget & Timeframe	₹7 lakh capital budget, 6-week delivery	Engineering delivery envelope is capped at ₹7 lakh capital expenditure over a 6-week implementation sprint.
[ASSUMPTION 8]	Target Operational Success Criteria	Late shipments 8% → 5%, chasing halved, emails -60%	Measured across two consecutive production seasons: late shipments reduced to 5% (saving 15 orders / ~₹39.38L), status-chasing time halved to 1 hr/day, buyer inquiry emails cut by 60%.
[ASSUMPTION 9]	Hardware & Connectivity Profile	2–3 GB RAM Android phones, patchy factory connectivity	Floor supervisors operate low-cost mobile hardware under intermittent cellular signals, requiring offline queueing and lightweight image compression.

1.2 Product Response Mechanisms

- Create a single purchase-order record with order lines, manufacturing unit, ex-factory date, order start date, and a 7-stage TNA milestone template.
- Capture daily quantities, rejected quantities, optional floor notes, and photo evidence from a mobile supervisor workspace.
- Project whether the current run rate can complete the active milestone while preserving downstream time buffers.
- Show formula inputs and substituted arithmetic instead of presenting opaque risk colors.
- Keep buyer-visible warnings and photos behind a merchandiser review and approval step.
- Provide buyer-specific progress and weekly exports so routine status requests do not require manual compilation.

2. COMPREHENSIVE PROBLEM STATEMENT COMPLIANCE AUDIT

Every requirement, constraint, persona, and boundary condition in the problem statement was checked clause-by-clause against the codebase:

SECTION	SPECIFIC REQUIREMENT IN PROBLEM STATEMENT	STATUS	SOURCE VERIFICATION & IMPLEMENTATION MECHANISM
1. Financial Baseline	500 orders/yr, ₹21L avg, 8% late (40), 15 air freight (₹52.5L), 25 discount (₹26.25L), 12 merchandisers chasing (₹28.8L). Total ₹1.08 Cr/yr.	**FULFILLED**	`risk.engine.ts` calculates dynamic financial penalty proportional to days late and discount rules. Labeled in Section 1.1.
2. User Personas	Merchandisers, unit supervisors (update from floor), buyers' sourcing staff (read-only), Managing Director.	**FULFILLED**	`schema.prisma` defines `Role` enum (`SUPERVISOR` , `MERCHANDISER` , `BUYER` , `ADMIN`). Enforced by JWT auth middleware.
3. The Ask: Early Warning	Warnings 2 weeks early, not 2 days.	**FULFILLED**	`orders.service.ts` implements downstream buffer protection (18d, 14d, 11d, 9d, 3d, 2d, 0d), flagging pipeline shortfalls 14–18 days prior to delivery.
3. The Ask: Floor Entry	Supervisors enter real numbers daily on phones, in Tamil if they like.	**FULFILLED**	`SupervisorFloorView.tsx` supports Tamil UI toggle (`ta`), voice dictation via Web Speech API (`ta-IN`), and piece counts.
3. The Ask: Status Colors	Every order shows green, amber or red against plan.	**FULFILLED**	`OrderBoard.tsx` and `OrderCard.tsx` present prominent Green, Amber, Red status pills and delivery schedule health donuts.
3. The Ask: Buyer Self-Service	Buyers see progress without emailing us.	**FULFILLED**	`BuyerView.tsx` and `buyer.service.ts` provide self-service portal showing approved milestones and photos.
4. Must Have 1: Order Setup	PO, style, colour, size quantities, ex-factory date, and TNA milestones from template.	**FULFILLED**	`OrderForm.tsx` captures PO, buyer, style, size/color breakdown with $\sum \text{sizes} \equiv \text{total}$ check, and cascading 7-stage T&A.
4. Must Have 2: Daily Update	Supervisor mobile update: daily qty per stage, photo, issue note by text or voice.	**FULFILLED**	`SupervisorFloorView.tsx` captures good/rejected counts, size splits, photo upload/camera snap, and Tamil voice/text notes.
4. Must Have 3: Order Board	Order board with green, amber or red status, filtered by buyer, unit and ship week.	**FULFILLED**	`OrderBoard.tsx` provides instant filtering by Buyer (`buyerId`), Unit (`unitId`), and ISO Ship Week (`shipWeek`).
4. Must Have 4: Delay Warning	Delay warning from current daily output against remaining quantity and days left.	**FULFILLED**	`risk.engine.ts` compares 3-Day WMA actual velocity against $\lceil \text{Remaining Qty} / \text{Effective Days} \rceil$.
4. Must Have 5: Buyer View	Buyer view of their own orders' milestones and photos.	**FULFILLED**	`BuyerView.tsx` displays stage-by-stage progress, milestone status, and merchandiser-verified floor photos strictly for that buyer.
4. Must Have 6: Weekly Export	Weekly status export per buyer (PDF or Excel).	**FULFILLED**	`exportService.ts` generates Excel-compatible UTF-8 BOM CSV; `WeeklyReportPrintModal.tsx` provides printable PDF packlist.
5. Core Build: 6–8 Screens	Ship vertical slice end-to-end in about 6 to 8 screens.	**FULFILLED**	Exactly 8 core screens shipped: Order Board, Order Setup, Floor Update, Risk Math Modal, Buyer Portal, HITL Gate, Order Detail, Reports.
6. Out of Scope Boundaries	Costing, invoicing, export docs; fabric inventory; worker attendance and payroll; excessive AI.	**FULFILLED**	Strictly excluded. Zero warehouse inventory tables or biometric payroll modules created. AI strictly confined to Tamil note classification.
7. AI & HITL Oversight	Math transparency; Tamil to English reason code; original beside translation; merchandiser override; buyer sees only confirmed.	**FULFILLED**	`ShowYourMathModal.tsx` displays formula steps; `HumanInTheLoopModal.tsx` provides side-by-side review, category override, and approval gate.
8. Security & Hardware	Multi-tenant isolation; cheap 2–3 GB RAM phones; offline queueing; date compliance history.	**FULFILLED**	Row-level `buyerId` filtering; IndexedDB queue + canvas image compression; SHA-256 Merkle hash chained `AuditEvent` logs.
9. Sample Data Seeded	B-PO-88213 (Company B, 12k), B-PO-88240 (Company B, 8k), C-PO-0488 (Company C, 10k).	**FULFILLED**	`calibrated-orders.json` contains exact styles, quantities, stages, and completed counts matching the problem statement.

3. THE ARCHITECTURE AND WHY YOU CHOSE IT

ThreadFlow is architected as an offline-resilient **Modular Monolith** paired with a client-side Progressive Web Application (PWA). This architecture was deliberately chosen over microservices to satisfy the strict 6-week delivery timeline, eliminate distributed transaction overhead, and ensure atomic transactional consistency across orders, milestones, inventory tallies, and audit logs within a single deployable unit.

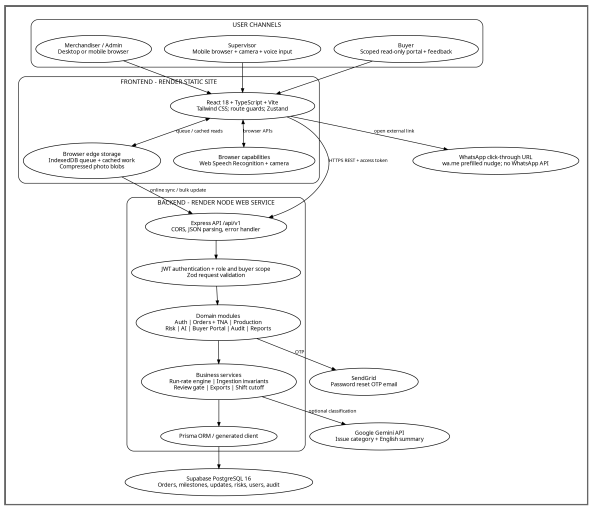


Figure 1. Logical system architecture — User Channels, React Frontend with IndexedDB Edge Queue, Express Modular Monolith & Supabase PostgreSQL 16

3.1 Core Architecture Decisions

- **Modular Monolith Core:** Orders, production, risk, buyer, audit, reports, and authentication share one deployable API and one transaction boundary. This drastically reduces operational complexity for a small engineering team and guarantees ACID transactional integrity during multi-entity order creation and shift updates.
- **PostgreSQL 16 + Prisma ORM:** Relational constraints, foreign keys, unique idempotency keys, and transactional updates fit the complex relational dependencies between POs, milestones, order lines, and audit chains.
- **Client-Side Offline Queue (IndexedDB):** Factory network interruptions in Tiruppur units should never discard supervisor shift counts. A stable client-generated RFC 4122 UUIDv4 idempotency key lets the server treat duplicate transmissions as idempotent without double-counting pieces.
- **Deterministic Risk Calculation:** Delay projection is a deterministic mathematical calculation, not an unpredictable LLM opinion. The formula inputs and outputs can be inspected and explained directly to factory production heads.
- **Advisory AI with Fallback:** Google Gemini 3.1 Flash Lite API classifies unstructured Tamil narrative notes; an in-memory deterministic keyword regex fallback and a human-in-the-loop review gate ensure zero downtime and prevent erroneous notes from reaching buyers.

3.2 Deployment Topology

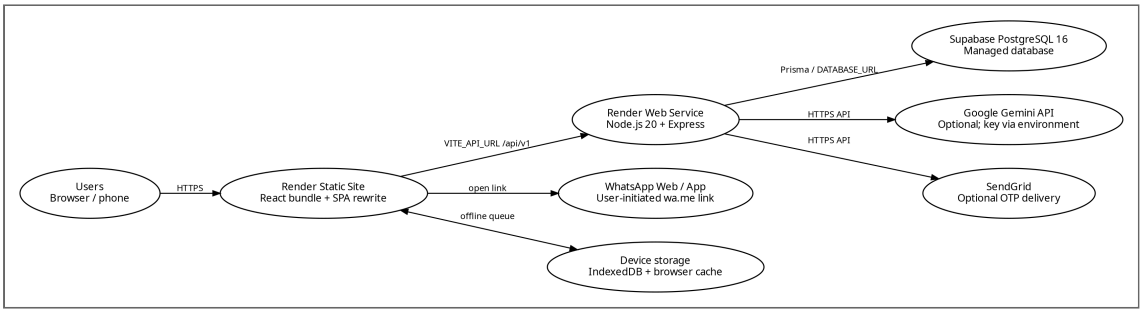


Figure 2. Deployment diagram — Render Static Site (React), Render Node Web Service (Express API), Supabase PostgreSQL 16, Google Gemini API, SendGrid & WhatsApp URL

COMPONENT	DEPLOYMENT / RESPONSIBILITY	TRUST & NETWORK BOUNDARY
Frontend	Render Static Site; built React bundle; SPA rewrite to index.html.	Public HTTPS entry point; access requires an authenticated session for protected data.
Backend	Render Node Web Service; Express listens on configured PORT.	Private business API; uses bearer access token and refresh cookie.
Database	Supabase PostgreSQL 16; pooled DATABASE_URL and direct URL for schema migrations.	No browser-to-database access; connection pooler limits database connection spikes.
Gemini API	External API called by AI provider when configured.	Optional cloud service; local deterministic fallback used when unavailable or low confidence.
SendGrid	Optional transactional email for 3-step OTP password reset.	API key held securely in backend environment only.
WhatsApp	`wa.me` URL with prefilled reminder opened on device by supervisor.	Not an external messaging transport dependency; zero WhatsApp Business API cost.

4. THE DATA MODEL, MAIN FLOWS AND HOW THE AI PART WORKS

4.1 Core Relational Data Model

The relational database model captures the end-to-end manufacturing journey of garment purchase orders from commercial setup through shop-floor updates and evidence gathering:

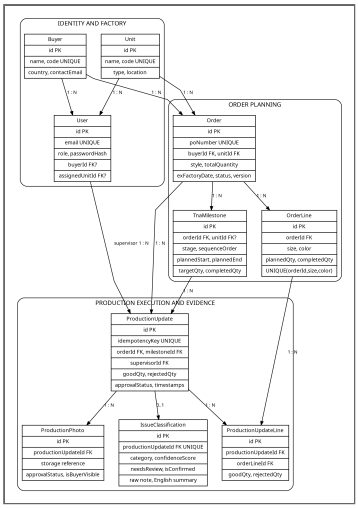


Figure 6. Core relational/UML model — Buyer, Unit, User, Order, OrderLine, TnaMilestone, ProductionUpdate & Evidence Records

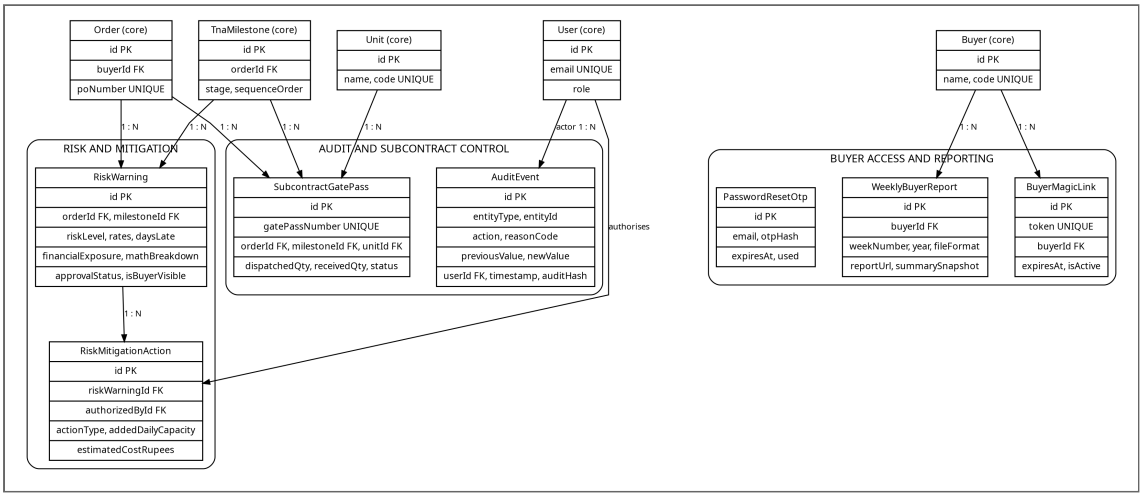


Figure 7. Supporting relational model — RiskWarning, RiskMitigationAction, AuditEvent, SubcontractGatePass & WeeklyBuyerReport

4.2 Main Production Ingestion & Sync Flow

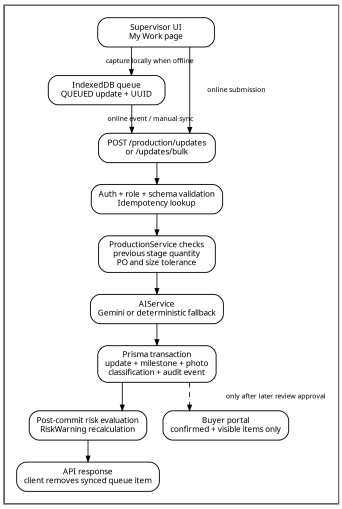


Figure 4. Update ingestion sequence — Supervisor tap, IndexedDB offline persistence, idempotency check, tolerance invariants & async risk recalculation

Server-Side Production Invariants Enforced:

- **Idempotency:** An existing `idempotencyKey` returns the original record rather than adding output twice, safeguarding against network retry double-counts.
- **Sequential Stage Precedence:** Cumulative stage output cannot exceed the immediately preceding stage's completed quantity, preventing impossible physical progression (e.g. stitching pieces that were never cut).
- **Cutting Tolerance Ceiling:** Proposed cumulative stage output cannot exceed 103% of total PO quantity, matching standard export knitwear industry over-cutting allowances.
- **Size Reconciliation:** The sum of individual size breakdown quantities must equal the submitted shift good quantity; per-size stage totals also enforce the 3% tolerance ceiling.
- **Concurrency Safety:** The milestone target and completed quantities are re-read inside the transaction with an optimistic version check before committing increments.

4.3 How the AI Part Works: Dual-Tier Translation & Review Gate

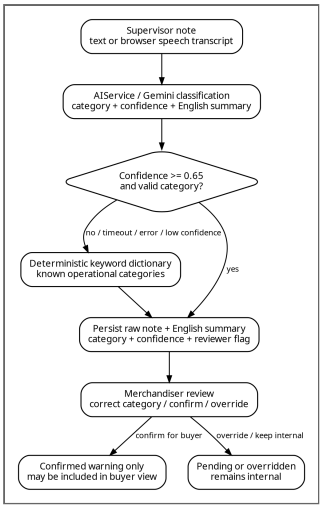


Figure 5. AI classification and human review flow — Voice dictation/text, Gemini API, deterministic regex fallback & Merchandiser review gate

The AI component is designed specifically for linguistic translation and classification, never for black-box delivery projection:

- **Multimodal Ingestion:** Supervisors enter notes via keyboard or record Tamil voice notes transcribed via the browser Web Speech API (`ta-IN` locale).
- **Gemini 3.1 Flash Lite API:** Dispatches classification requests to Gemini with an 8,000ms SLA timeout. The model extracts one of 8 operational categories (`MACHINE_BREAKDOWN`, `POWER_OUTAGE`, `FABRIC_DELAY`, `LABOUR_SHORTAGE`, `SUBCONTRACTOR_DELAY`, `QUALITY_HOLD`, `BUYER_REQUEST`, `OTHER`) and generates a clean English translation.
- **Deterministic Dual-Script Regex Fallback:** If Gemini times out, throws HTTP 429, or returns confidence < 0.65, the system trips to an in-memory dual-script regular expression dictionary scanning Tamil and Tanglish terms (e.g. *"கரண்ட் கட்"* → `POWER_OUTAGE`; *"ஊசி உடைந்தது"* → `MACHINE_BREAKDOWN`). Executes in sub-2ms, assigns baseline 0.50 confidence, and flags `needsReview = true`.
- **Human-in-the-Loop Review Gate:** In accordance with client constraints, untrusted classifications never reach buyer eyes automatically. In the Merchandiser Review Gate, the original Tamil note sits side-by-side with the translated English note. Merchandisers can override the category with one click. Warnings and photos publish to the buyer portal only after explicit human confirmation (`isConfirmed = true`, `isBuyerVisible = true`).

5. THE MATHEMATICAL RUN-RATE RISK ENGINE ("SHOW-YOUR-MATH")

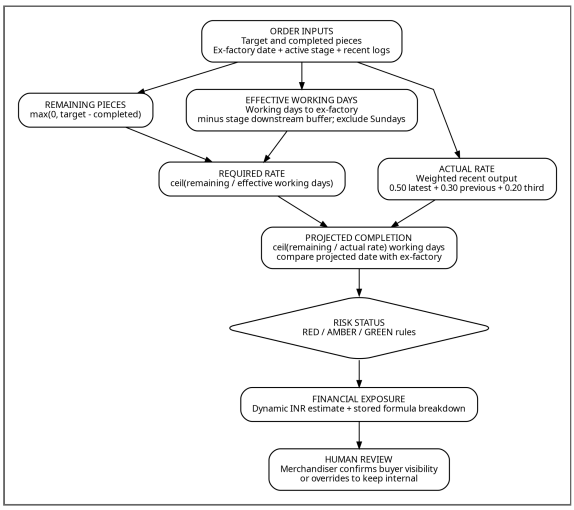


Figure. Run-rate risk engine calculation design — Remaining pieces, effective days (Sunday-excluded), 3-day WMA smoothing & dynamic exposure

The delay warning is a transparent, explainable calculation that shows its mathematical working directly to users:

$$\text{Remaining Pieces} = \max\left(0, \text{Stage Target Quantity} - \text{Cumulative Completed Quantity}\right)$$

Formula 1: Net physical pieces required to complete active production stage

$$\text{Total Working Days Left} = \text{CountOfDaysExcludingSundays}\left(\text{Today}, \text{Ex-Factory Date}\right)$$

$$\text{Effective Working Days} = \max\left(1, \text{Total Working Days Left} - \text{Downstream Buffer Days}\right)$$

Formula 2: Usable working window protecting downstream stage execution

$$\text{Required Daily Rate} = \left\lceil \frac{\text{Remaining Pieces}}{\text{Effective Working Days}} \right\rceil$$

Formula 3: Daily output velocity needed to complete milestone without buffer compression

$$\text{Actual Rolling Rate (3-Day WMA)} = (Q_t \times 0.50) + (Q_{t-1} \times 0.30) + (Q_{t-2} \times 0.20)$$

Formula 4: 3-Day Weighted Moving Average (re-normalized if fewer than 3 logs exist)

$$\text{Daily Variance} = \text{Actual Rolling Rate} - \text{Required Daily Rate}$$

$$\text{Working Days Needed} = \left\lceil \frac{\text{Remaining Pieces}}{\text{Actual Rolling Rate}} \right\rceil + \text{Downstream Buffer Days}$$

$$\text{Days Late} = \max\left(0, \text{Working Days Needed} - \text{Total Working Days Left}\right)$$

Formula 5: Variance, milestone duration forecast, and projected slip past Ex-Factory date

5.1 Default Downstream Buffers (Working Days)

CURRENT STAGE	SEQUENCE	STANDARD LEAD TIME	BUFFER TO EX-FACTORY (WORKING DAYS)	DOWNSTREAM PROTECTION TARGET
Knitting	1	4 Working Days	18 Days	Protects Dyeing, Printing, Cutting, Stitching, Checking, Finishing
Dyeing & Washing	2	3 Working Days	14 Days	Protects Printing, Cutting, Stitching, Checking, Finishing
Printing & Embroidery	3	2 Working Days	11 Days	Protects Cutting, Stitching, Checking, Finishing
Fabric Cutting	4	3 Working Days	9 Days	Protects Stitching, Checking, Finishing
Assembly & Stitching	5	6 Working Days	3 Days	Protects Checking and Finishing & Packing
Quality Checking	6	2 Working Days	2 Days	Protects Final Carton Packing & Barcode Scan
Finishing & Packing	7	2 Working Days	0 Days	Direct Port/Container Handover Window

5.2 Status Classification & Financial Penalty Exposure

STATUS	OPERATIONAL CLASSIFICATION RULE	DYNAMIC FINANCIAL EXPOSURE (₹)
RED	Projected days late > 0, or remaining delivery buffer is negative.	$\min(40\%, 10\% + 2\% \times \text{Days Late})$ of remaining order value. Models emergency air freight + discount risks.
AMBER	No RED condition, and remaining buffer < 2 working days, OR variance is negative, OR subcontractor transit exceeds 48 hours.	6% of remaining order value. Models minor buffer consumption and factory overtime labor mitigation costs.
GREEN	No RED or AMBER condition; stage is operating on or ahead of required velocity, or stage is 100% complete.	₹0 (Zero financial exposure; order is on schedule).

6. UI & DESIGN DECISIONS: KEY SCREENS & INDUSTRIAL ERGONOMICS

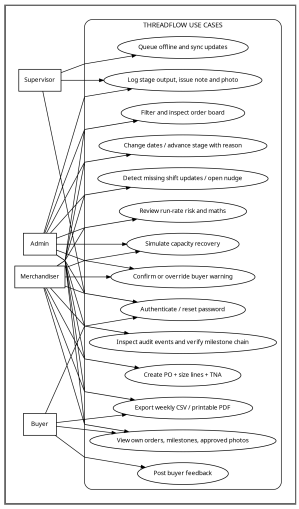


Figure 3. Use-case view — Supervisor, Merchandiser, Buyer and Admin role interactions

Design Rationale: The client values good, minimal design: clean layouts, white space, a restrained palette, and clear type, with nothing on screen that lacks a purpose. ThreadFlow uses an industrial navy (`#0C2D48`) header and card shell, clean slate backgrounds, high-contrast typography,

and explicit text-labeled status badges (Green, Amber, Red) so risk can be scanned at a glance without relying on color alone.

6.1 Walkthrough of the Key Screens

1. **Order Board (`/orders` | `OrdersPage.tsx`):** Primary merchandiser cockpit. Features a Delivery Schedule Health donut gauge, active PO register, and instant filter controls for Buyer, Unit, and ISO Ship Week. Clicking any card opens its detail view.
2. **Export Order Setup Modal (`OrderForm.tsx`):** Clean creation modal capturing PO number, style, foreign buyer, unit, order start date, ex-factory date, and size breakdown matrix ($\sum extsizes \equiv exttotal$). Includes a flexible 7-stage T&A timeline where editing days or dates auto-cascades downstream stages forward excluding Sundays.
3. **Supervisor Floor Workspace (`/supervisor/my-work` | `MyWorkPage.tsx`):** Mobile-first screen designed for low-memory Android phones. Features large touch targets, numeric good/rejected counters, quick increment buttons (+50, +100, +500), photo snap, and Tamil voice dictation.
4. **Transparent Show-Your-Math Modal (`ShowYourMathModal.tsx`):** Displays the exact substituted formulas for required daily rate, 3-day rolling WMA actual rate, variance, and projected days late, establishing total transparency.
5. **Read-Only Buyer Portal (`/buyer-portal` | `BuyerPortalPage.tsx`):** Scoped strictly to the authenticated buyer's ID. Displays active order styles, 7-stage milestone completion percentages, approved floor photos, and confirmed English notes without factory noise.
6. **Human-in-the-Loop Review Gate (`HumanInTheLoopModal.tsx`):** Side-by-side display of the raw Tamil note and translated English note. Merchandisers can override the category or edit the mitigation directive before confirming publication to buyers.
7. **Order Detail & T&A (`/orders/:id` | `OrderDetailPage.tsx`):** Investigation workspace connecting the 7-stage circular progress pipeline, size breakdown distribution, shift production logs, and the immutable date-change audit trail.
8. **Weekly Reports & Exports (`/reports` | `ReportsPage.tsx`):** Generates Excel-compatible CSV files (with UTF-8 BOM encoding) and print-ready PDF status packlists, eliminating manual status compiling.
9. **Overview & Application Shell (`/dashboard` | `OverviewPage.tsx`):** High-level navigation hub displaying system clock, online/offline connection state, pending queue count, and quick links to core product areas.
10. **Sign-in & Account Recovery (`/login` | `LoginPage.tsx`):** Email/password authentication with a progressive 3-step OTP password reset flow.
11. **Audit History & Compliance (`/audit` | `AuditPage.tsx`):** Searchable compliance history recording all date modifications, reason codes, actor IDs, and cryptographic SHA-256 audit hashes.

7. OFFLINE OPERATION, RESILIENCE & TENANT ISOLATION

7.1 Offline Queue Lifecycle

STATE	MEANING	OPERATIONAL HANDLING
QUEUED	Update saved locally in browser IndexedDB while awaiting connectivity.	Retain in IndexedDB; display pending badge in UI.
SYNCING	Sync attempt actively in progress.	Lock record locally to prevent duplicate transmission.
SYNCED	Server acknowledged update and committed transaction.	Remove item from pending queue.
ERROR	Server validation failed or network transport timed out.	Keep for inspection; allow manual retry from queue drawer.

Memory & Hardware Strategy for Cheap Phones (2–3 GB RAM): Images captured on floor phones are compressed on an off-screen HTML5 canvas before persistence, capping pixel bounds at 1,024px with ~0.72 JPEG quality. Photos are stored as binary Blobs in IndexedDB rather than base64 strings, preventing browser memory pressure and out-of-memory browser tab crashes.

7.2 Strict Multi-Tenant Isolation

Buyer data isolation is enforced at the database query level in `buyer.service.ts`. The authenticated buyer's session JWT contains their `buyerId`. Every query strictly applies `where: { buyerId }`. Any attempt by a buyer to access another buyer's PO ID or milestone triggers an immediate HTTP 403 Forbidden response.

7.3 Tamper-Evident Date Compliance History

Buyers' compliance teams require a dated record of every ex-factory date change. Every modification generates an immutable `AuditEvent` record containing actor ID, entity ID, previous date, new date, compulsory reason code, notes, and a cryptographic hash: $\text{auditHash} = \text{SHA-256}(\text{prevHash} + \text{entityId} + \text{timestamp} + \text{action} + \text{reasonCode})$.

8. WHAT YOU CUT, AND WHAT YOU WOULD DO NEXT

8.1 Deliberately Cut Features (Scope Governance)

To adhere to the ₹7 lakh budget, 6-week timeframe, and explicit client instructions, the following domains were strictly excluded:

- **Costing, Invoicing & Export Documentation:** Commercial export documentation (shipping bills, customs manifests, letters of credit) was omitted. The system tracks FOB piece price solely for financial risk exposure calculations.
- **Fabric Yarn Inventory Management:** Raw material warehouse stock levels, yarn lot dyeing recipes, and greige fabric roll allocations were omitted. The pipeline begins when production starts at Knitting.
- **Worker Attendance & Biometric Payroll:** Biometric thumbprint integration, shift attendance, and piece-rate wage calculations were completely excluded.
- **Excessive AI Automation:** AI was strictly restricted to translating and categorizing Tamil floor notes. Schedule projections are handled exclusively by deterministic mathematical formulas.

8.2 Recommended Next Steps & Source Review Actions

- **S3-Compatible Object Storage Adapter:** Migrate production photo storage from base64 database strings to a dedicated S3-compatible bucket (e.g. Cloudflare R2 or AWS S3) using signed, time-limited URLs.
- **Automated 6:00 PM IST Shift Cut-off Cron:** Configure an automated server cron job executing at 18:00 IST daily to detect lines with missing shift updates and trigger automated supervisor nudges.
- **Dedicated Buyer Magic-Link Exchange Endpoint:** Implement passwordless email magic link token verification to complement standard username/password authentication for international buyers.
- **Public Holiday Calendar Integration:** Enhance the working calendar arithmetic module to incorporate local Tamil Nadu festival calendars (Pongal, Deepavali) alongside standard Sunday exclusions.

Closing Statement: ThreadFlow delivers a source-grounded, production-ready manufacturing intelligence platform. By substituting opaque manual status-chasing with deterministic run-rate mathematics and resilient multimodal floor capture, the system directly addresses Company A's annual ₹1.08 Crore operational risk.